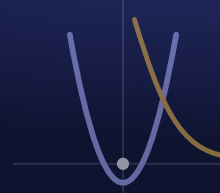


Principles of Applied Engineering

TEKS §127.781 • CTE Level 1 • Grade 9 • Mr. Gavaldon • Da Vinci School

robertgavaldon@burnhamwood.org


 SEMESTER 1
Week 2

MONDAY

Day 6 - Launch: Engineering Firms + Pick Your Problem

Aug 17

OBJECTIVE

Students will form engineering firms, choose a real problem, and define why it matters (Design Brief p.1).

TEKS

§127.781(d)(2)(G) engineering design process – planning; (d)(3)(C) sketching the design

ELPS / EB SUPPORT

Design brief and planning packet give sentence frames plus a labeled force diagram (VC, DP, CD). Pre-teach constraint / criteria / rolling-resistance from a picture bank (PV). Mixed teams with builder / recorder / tester roles; native-language partner talk before writing the justification (PN, WT).

AGENDA • 45 MIN

6' Bell-work (format + example): name a badly-designed everyday thing

10' Launch: the firm mission + week map

8' Worked example: a firm end-to-end

6' Randomize firms + assign 4 roles

15' Pick your problem + start Design Brief page 1

SLIDES / ACTIVITY

Engineering Firms – Launch

NOTEBOOK

Glue in the car planning sheet. Sketch your design and label the body, axles, and wheels.

TURN IN

Engineering Firms - Design Brief packet, page 1 (firm, roles, problem, why)

TUESDAY

Day 7 - Define the Problem

Aug 18

OBJECTIVE

Students will define their problem with a stakeholder, evidence, testable requirements, and a success statement (Design Brief p.2).

TEKS

§127.781(d)(2)(G) build the prototype (design process); (d)(4) safe, responsible use of tools & materials

ELPS / EB SUPPORT

Teacher models each assembly step at the front and leaves an annotated photo sequence up (VC, DP, RS). Build-log terms (tolerance, clearance, parallel) pre-taught with a picture bank (PV, CD). Roles keep every student hands-on; partner talk before recording corrective action (PN, WT).

AGENDA • 45 MIN

6' Bell-work: who has it + how you know

12' Define like an engineer: who, evidence, requirements, success

10' Worked example + vocab

17' Write 3-5 testable requirements - Design Brief page 2

SLIDES / ACTIVITY

Engineering Firms – Define It

NOTEBOOK

Log your build: what went well, what didn't, and the one fix you'll make on test day.

TURN IN

Engineering Firms - Design Brief packet, page 2 (definition + requirements)

OBJECTIVE

Students will sketch multiple solutions, choose one against their requirements, and generate an AI product image (Design Brief p.3).

TEKS

§127.781(d)(2)(G) test & evaluate (design process); (d)(3)(D) present results with visualization

ELPS / EB SUPPORT

Data sheet is mostly numbers with one sentence of reasoning (low language load, CD, DP). Pre-teach independent/dependent/controlled variable and mean/range with a picture bank (PV). Pair-share the analysis before writing the conclusion (WT, PN).

AGENDA • 45 MIN

6' Bell-work: why sketch more than one idea

10' Sketch 3 solutions + choose with a reason/trade-off

10' Write the product-image prompt + generate

19' Finish Design Brief page 3 (sketches, choice, image)

SLIDES / ACTIVITY

Engineering Firms – Design It

NOTEBOOK

Record both trial distances and the single change you made. Was your prediction right?

TURN IN

Engineering Firms - Design Brief packet, page 3 (design + AI image)

OBJECTIVE

Students will build a 5-beat pitch deck with AI and prepare to defend it under questioning (Design Brief p.4).

TEKS

§127.781(d)(3)(C) computer-aided design & drafting (CADD); (d)(1)(A) professional online conduct

ELPS / EB SUPPORT

Video has a guided sheet with sentence stems and target terms to catch (RS, CD, DP). Pre-teach CAD / parametric / extrude with labeled screenshots (PV, VC). 'Show me your email' is a low-stakes one-on-one check with wait time (WT); native-language partner support at the station (PN).

AGENDA • 45 MIN

6' Bell-work: what makes a pitch believable

10' The 5 beats + AI-build the deck from your packet

12' Drill the defense: assign who owns each question

17' Finish deck + one full practice run - Design Brief page 4

SLIDES / ACTIVITY

Engineering Firms – Build the Pitch

NOTEBOOK

Note 3 things Fusion can do, and one thing you want to build in it this year.

TURN IN

Engineering Firms - Design Brief packet, page 4 (pitch deck + Q&A prep)

OBJECTIVE

Student firms will pitch their product and defend it under Q&A from a review panel, scored on a shared rubric.

TEKS

§127.781(d)(3)(C) set up & navigate CAD (CADD); (d)(1)(B) collaborate & support peers

ELPS / EB SUPPORT

Account and modeling steps are numbered and screenshotted in Classroom (VC, DP, CD). Pre-teach interface terms (sketch, extrude, ViewCube, timeline) with labeled images (PV). Finishers pair with peers using an 'ask three, then me' protocol; native-language walkthrough allowed (PN, WT).

AGENDA • 45 MIN

6' Bell-work: what a great audience does

6' Format + rubric (how you are scored)

33' Design Review: 3-min pitch + 2-min defense per firm

5' Exit ticket + Week 3 (30 Days of Fusion) tease

SLIDES / ACTIVITY

Engineering Firms – Design Review

NOTEBOOK

Paste your Fusion account email (not password). Note where the Sketch button and timeline are.

TURN IN

Engineering Firms - Design Review rubric (25 pts) + reflection

